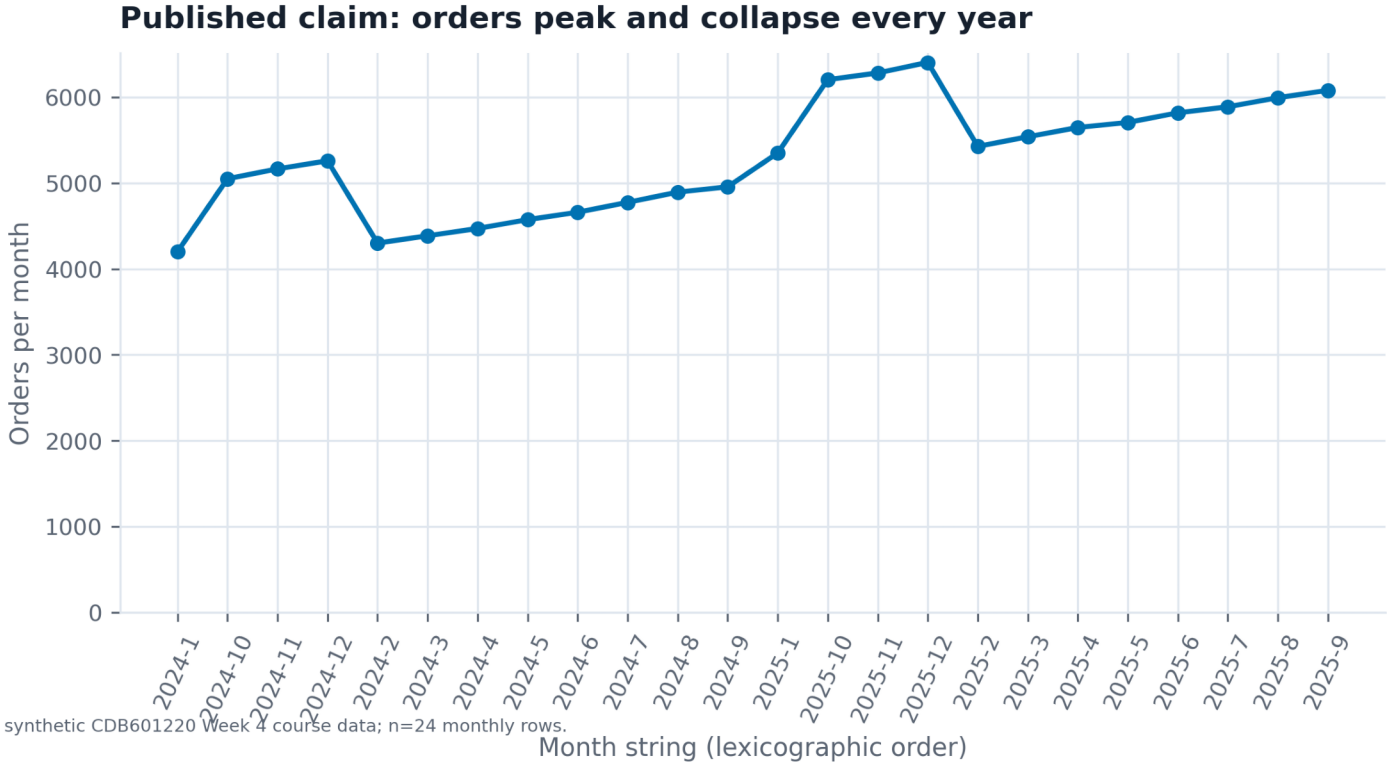
Case 1 — Published recipe



Source: synthetic CDB601220 Week 4 course data; n=24 monthly rows.

Case 1 — Redesign

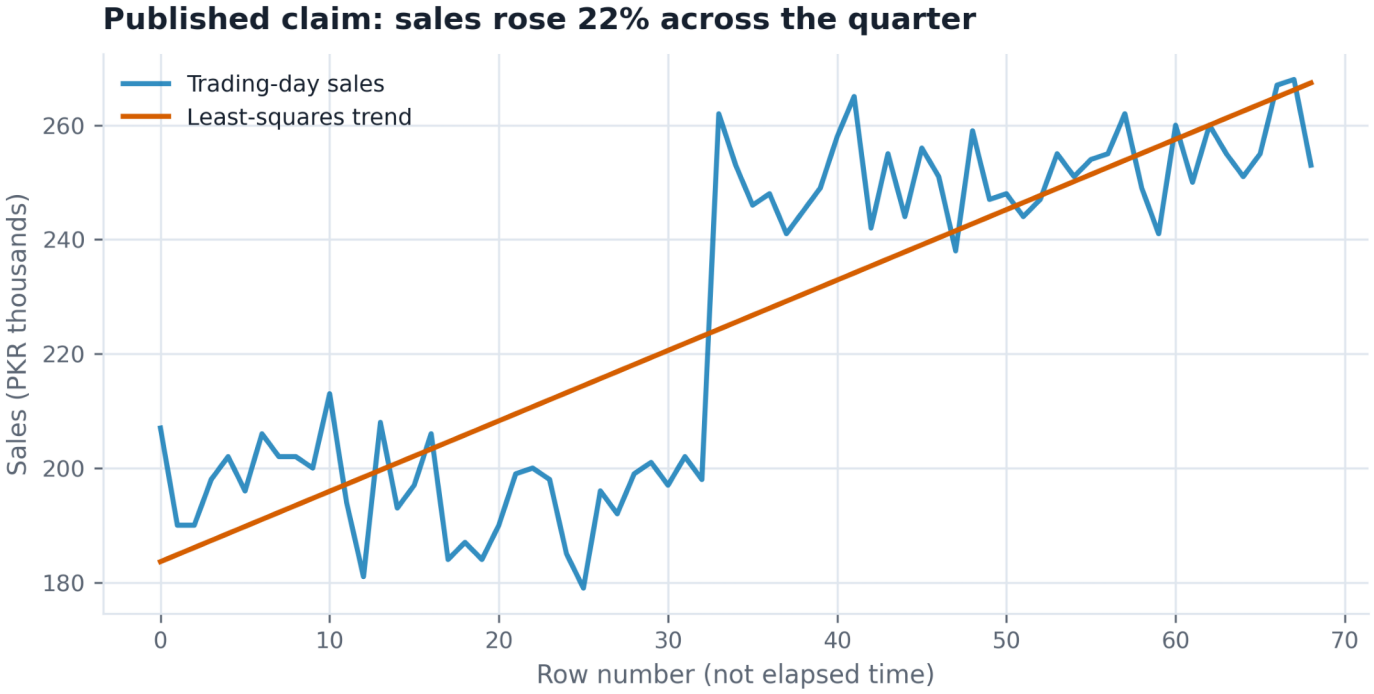


Source: synthetic CDB601220 Week 4 course data; YYYY-M parsed as year-month.

Case 1: diagnosis

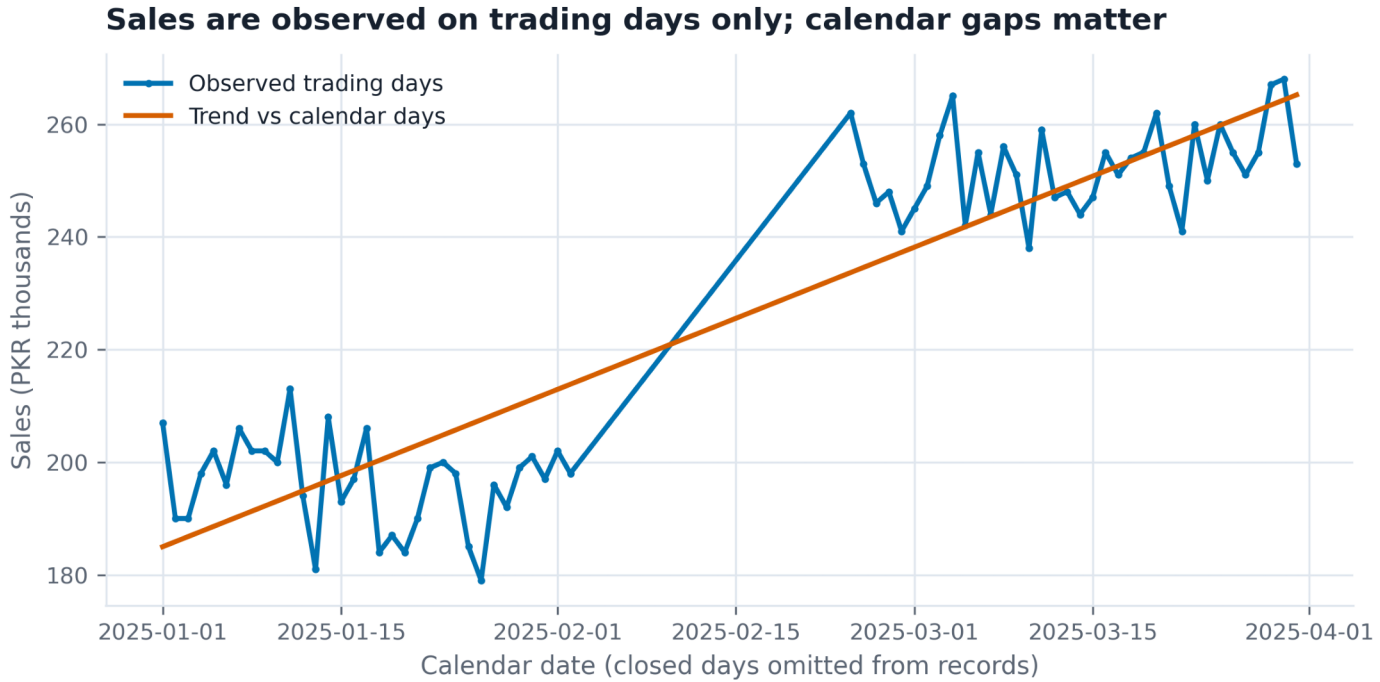
Mechanism: `YYYY-M` is non-zero-padded text, so lexical sorting orders month 10/11/12 before month 2. The false drops are 958
and 978 orders; parsing to datetime restores chronological order. First objection ‘seasonality is not proven’ is incomplete: the specific visible collapses are manufactured by string sorting, not annual cycles.

Case 2 — Published recipe



Source: synthetic CDB601220 Week 4 course data; n=69 trading-day rows.

Case 2 — Redesign

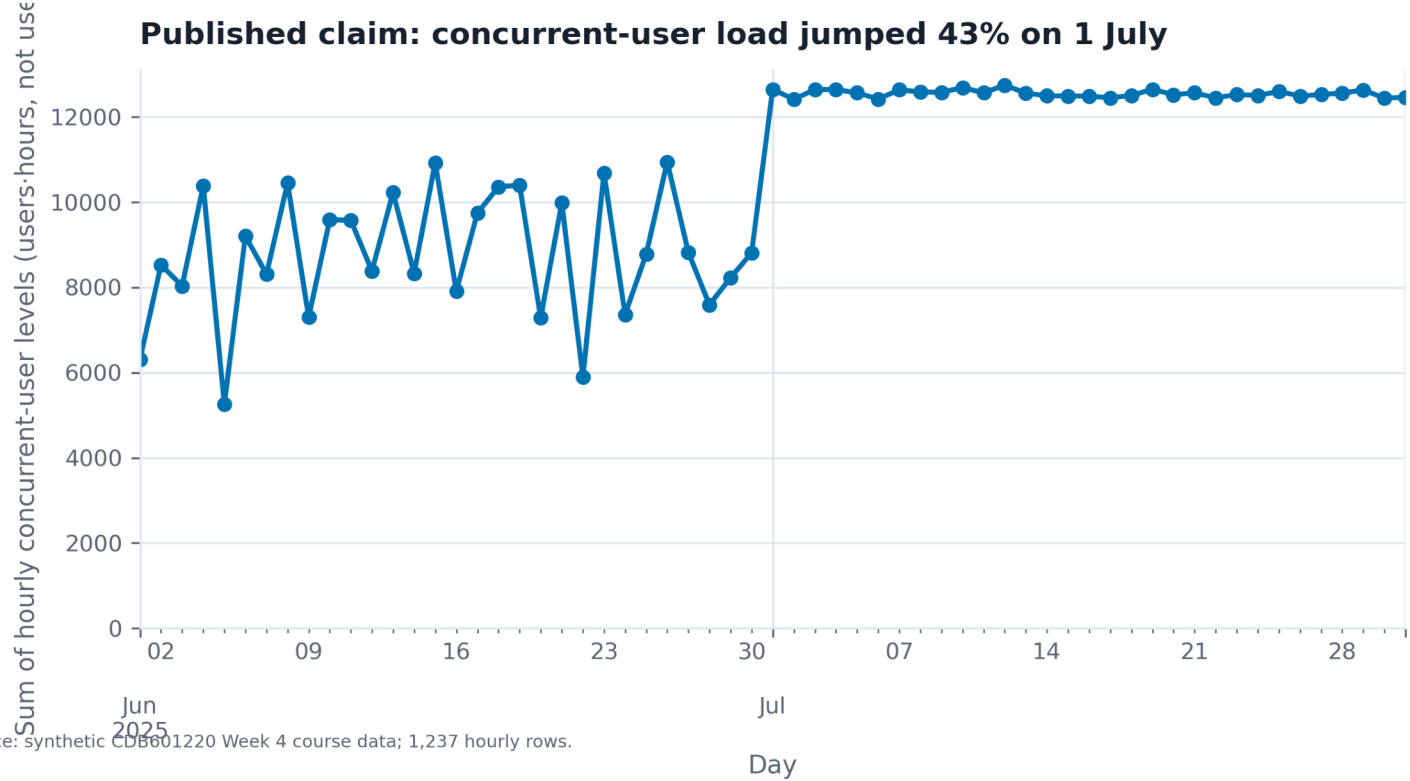


Source: synthetic CDB601220 Week 4 course data; n=69 observed trading days; line connects observations across closed days.

Case 2: diagnosis

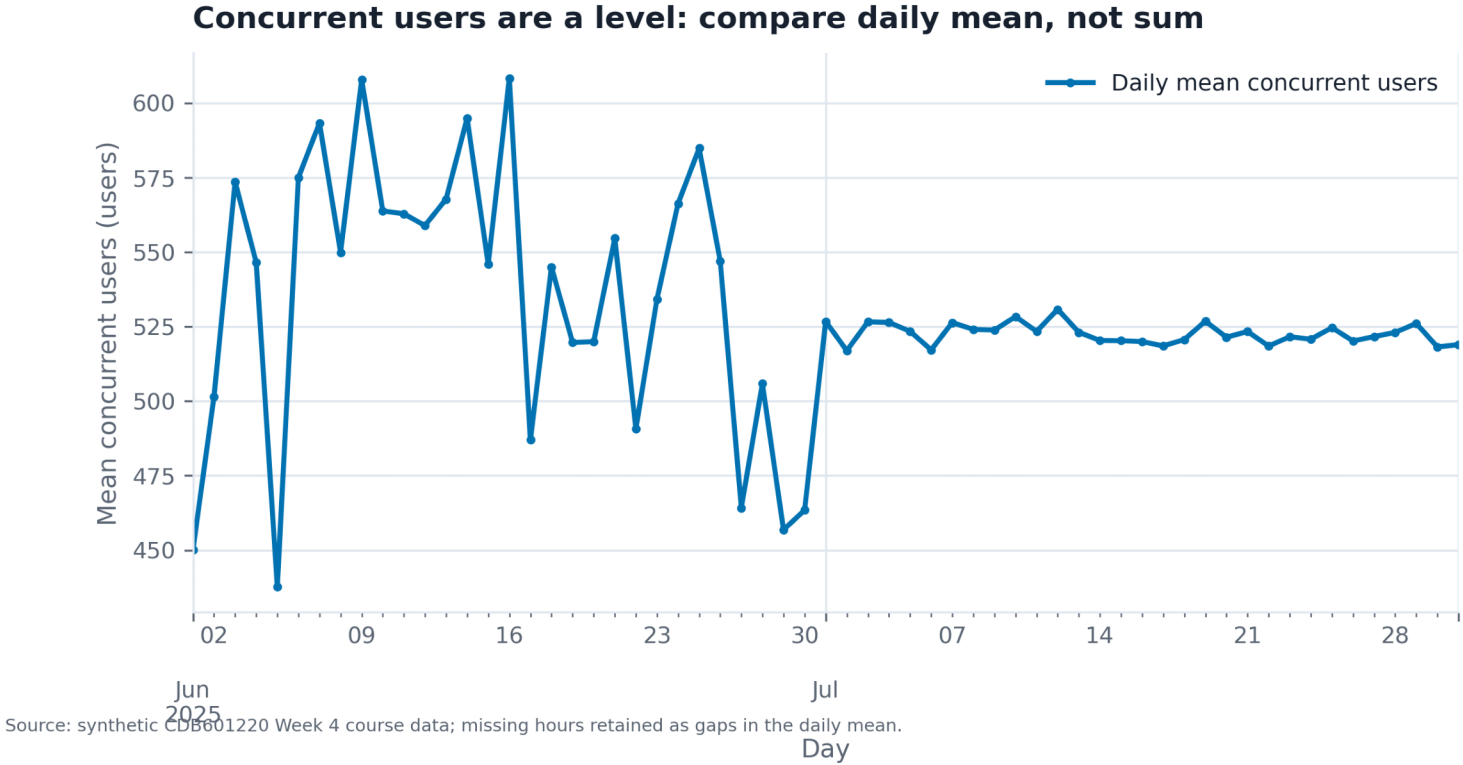
Mechanism: row number treats 69 trading observations as equally spaced despite 21 missing calendar dates (90 calendar days in
span). The 207→253 endpoint rise is +22.2%, but it is not evidence of steady daily growth across elapsed time; sales fluctuate and the row index conceals closures. The decoy ‘sample too small’ is not supported: n=69 is the stated dataset; the
problem is the x-coordinate and missing non-trading dates.

Case 3 — Published recipe



Source: synthetic CDB601220 Week 4 course data; 1,237 hourly rows.

Case 3 — Redesign

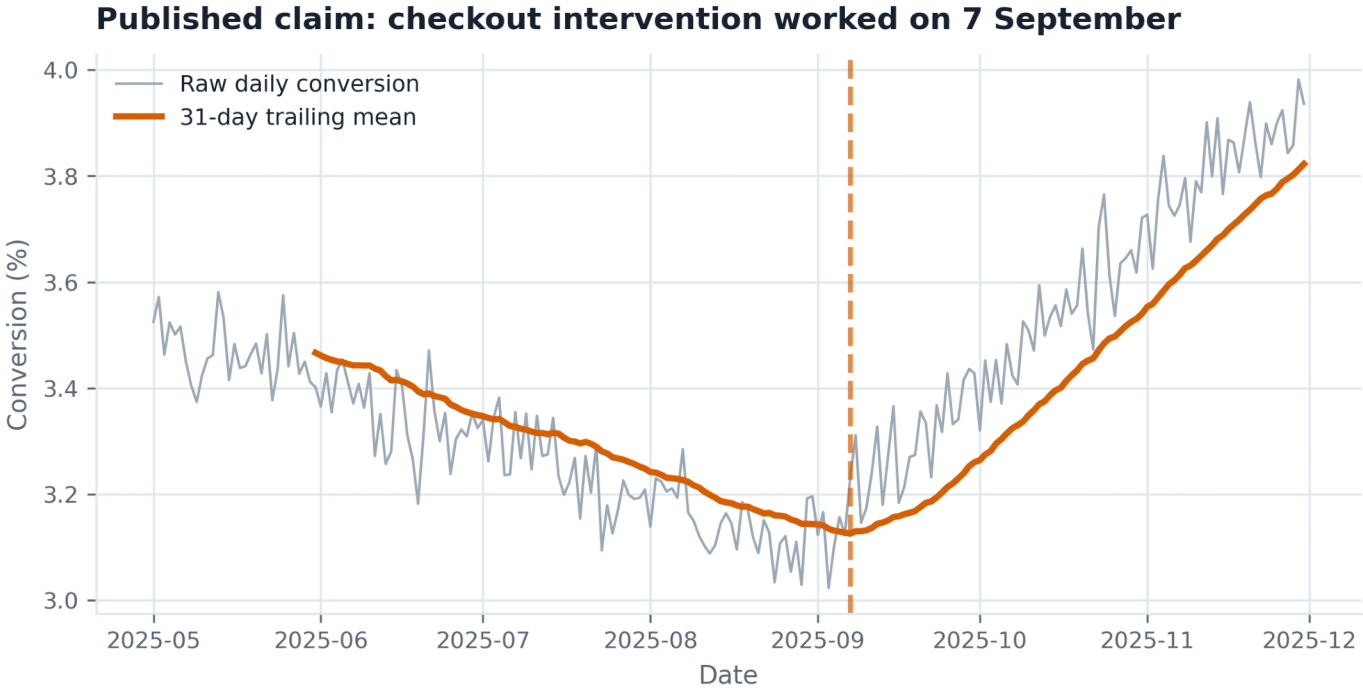


Source: synthetic CDB601220 Week 4 course data; missing hours retained as gaps in the daily mean.

Case 3: diagnosis

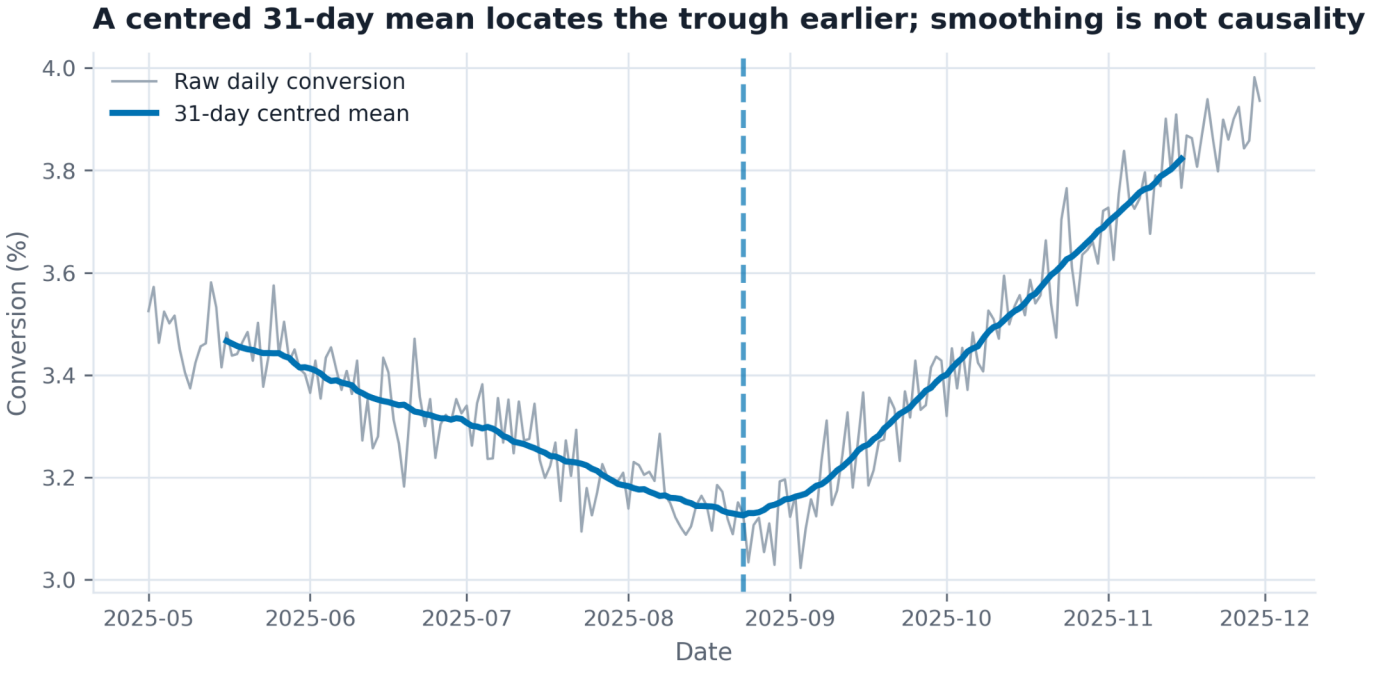
Mechanism: `concurrent_users` is an instantaneous level, not a flow. Summing it makes the result depend on hours recorded per
day. The index has 0 duplicate timestamps and 227 missing hourly timestamps; therefore coverage also varies. Correct daily means by month are 535.9 in June and 522.6 in July (-2.5%), not a justified 43% level jump. The decoy ‘July has more users because sum is larger’ confuses a level with an accumulated flow.

Case 4 — Published recipe



Source: synthetic CDB601220 Week 4 course data; 214 daily rows.

Case 4 — Redesign

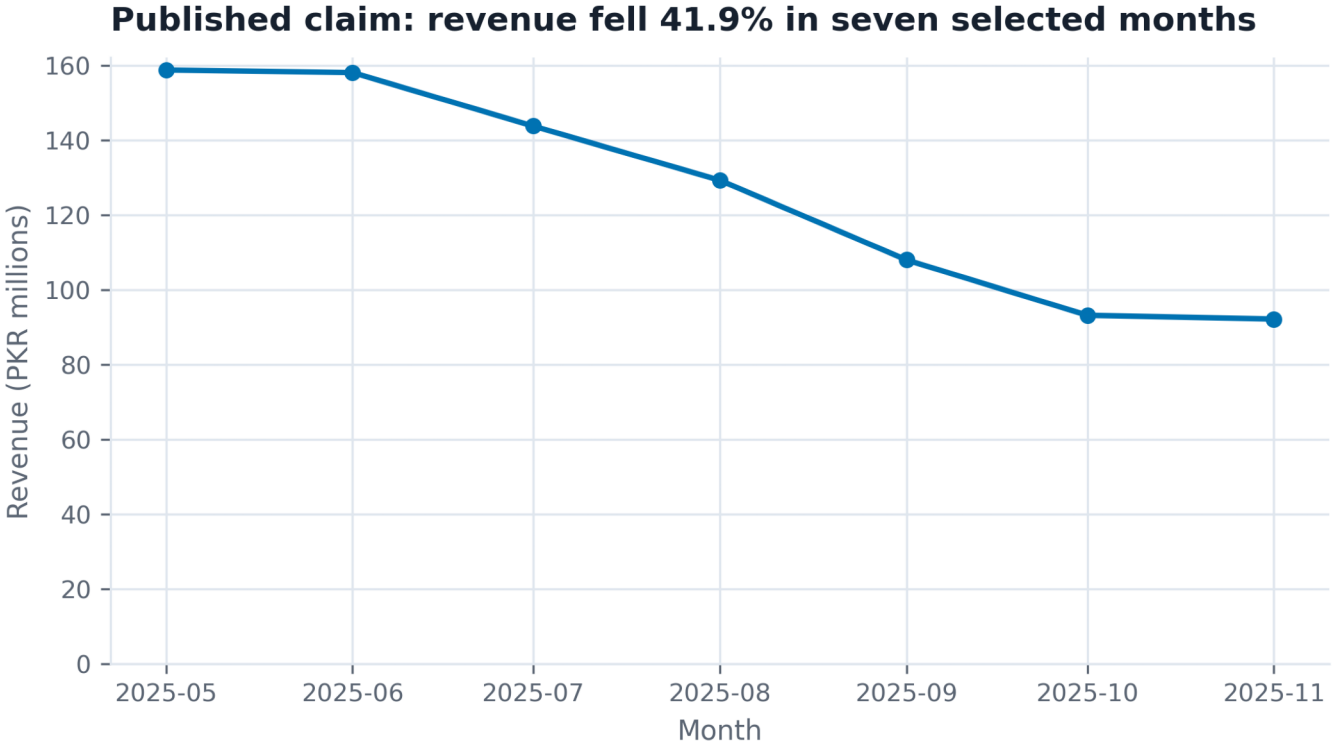


Source: synthetic CDB601220 Week 4 course data; centred mean is retrospective and uses future observations.

Case 4: diagnosis

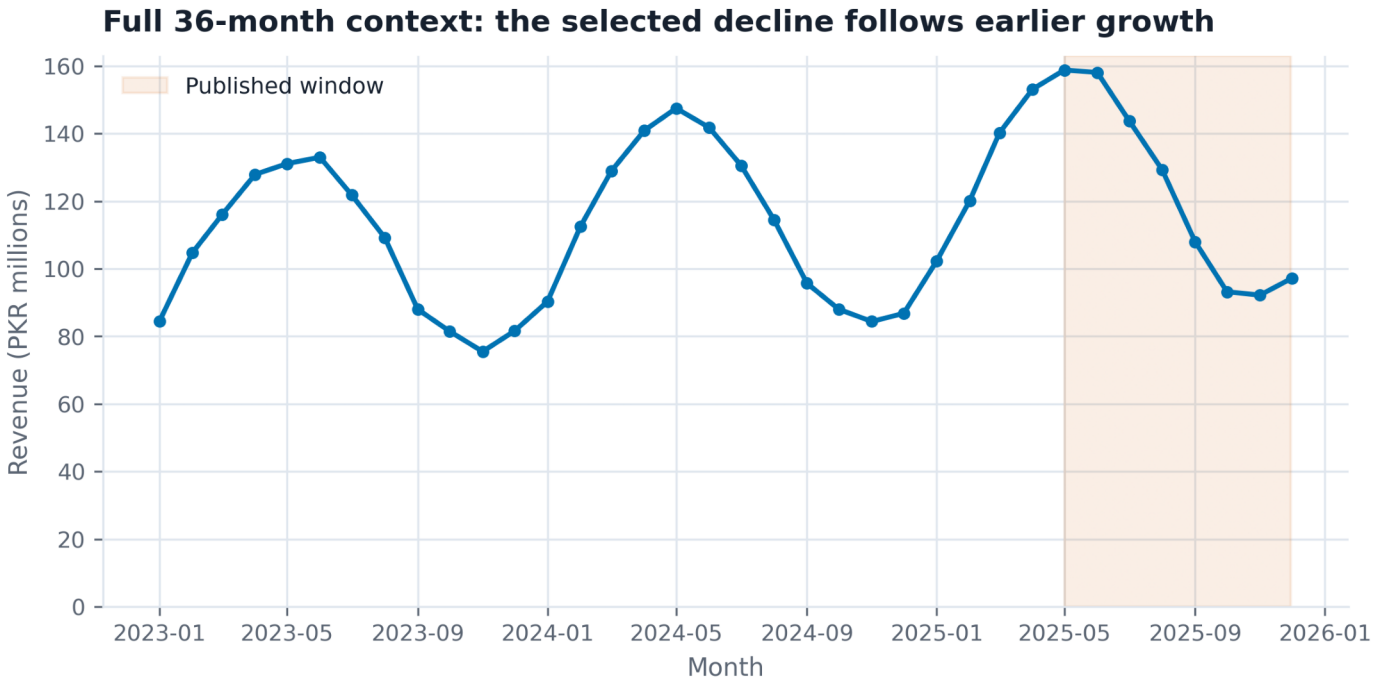
Mechanism: a 31-day trailing mean lags a turn because it only includes the current and previous 30 days. Trailing minimum is 2025-09-07, while centred-window minimum is 2025-08-23, a 15-day difference. This is a window-induced lag, not proof that the
intervention began working on the smoothed minimum date. The raw series and a centred view show timing uncertainty; causal
attribution needs a counterfactual.

Case 5 — Published recipe



Source: synthetic CDB601220 Week 4 course data; selected May–November 2025 window, n=7.

Case 5 — Redesign

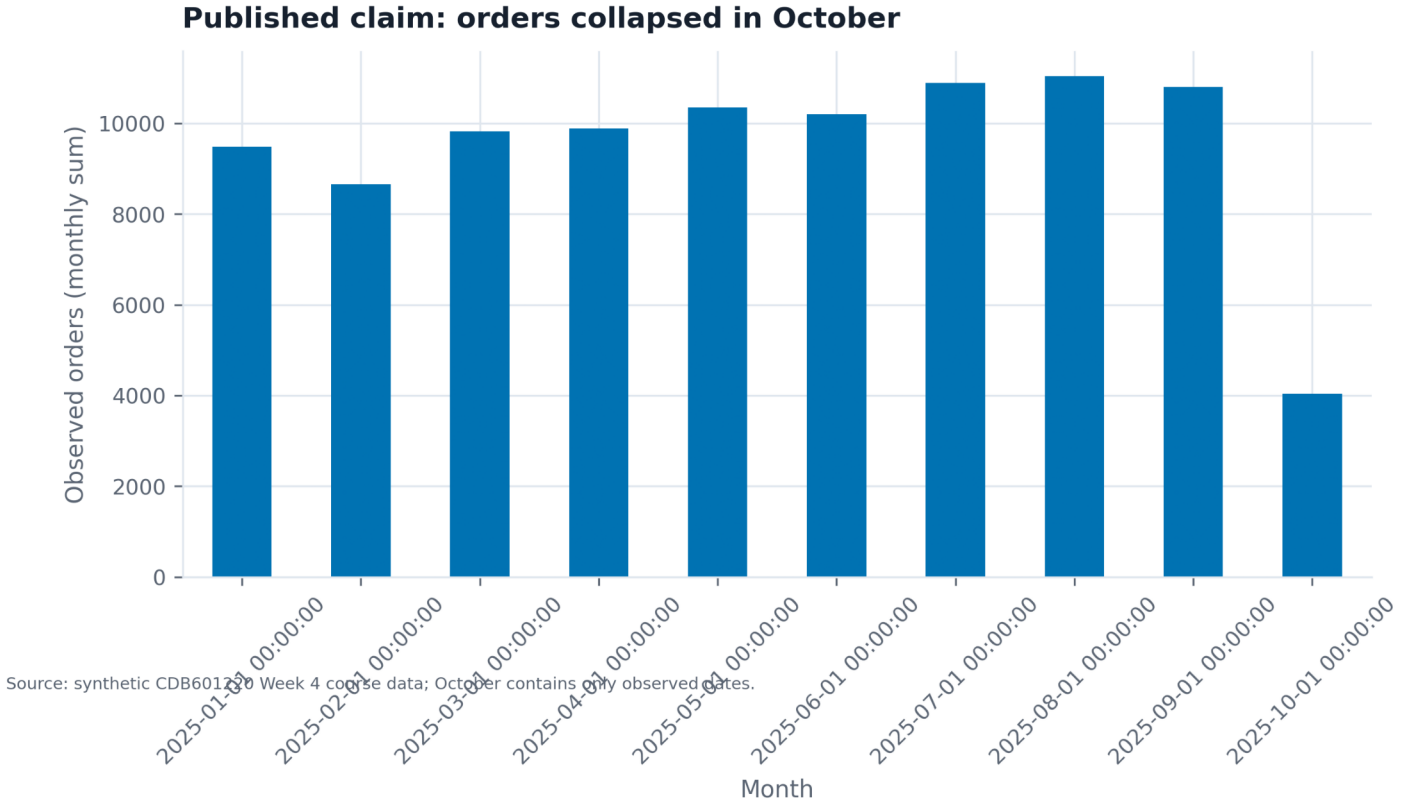


Source: synthetic CDB601220 Week 4 course data; complete monthly series, n=36.

Case 5: diagnosis

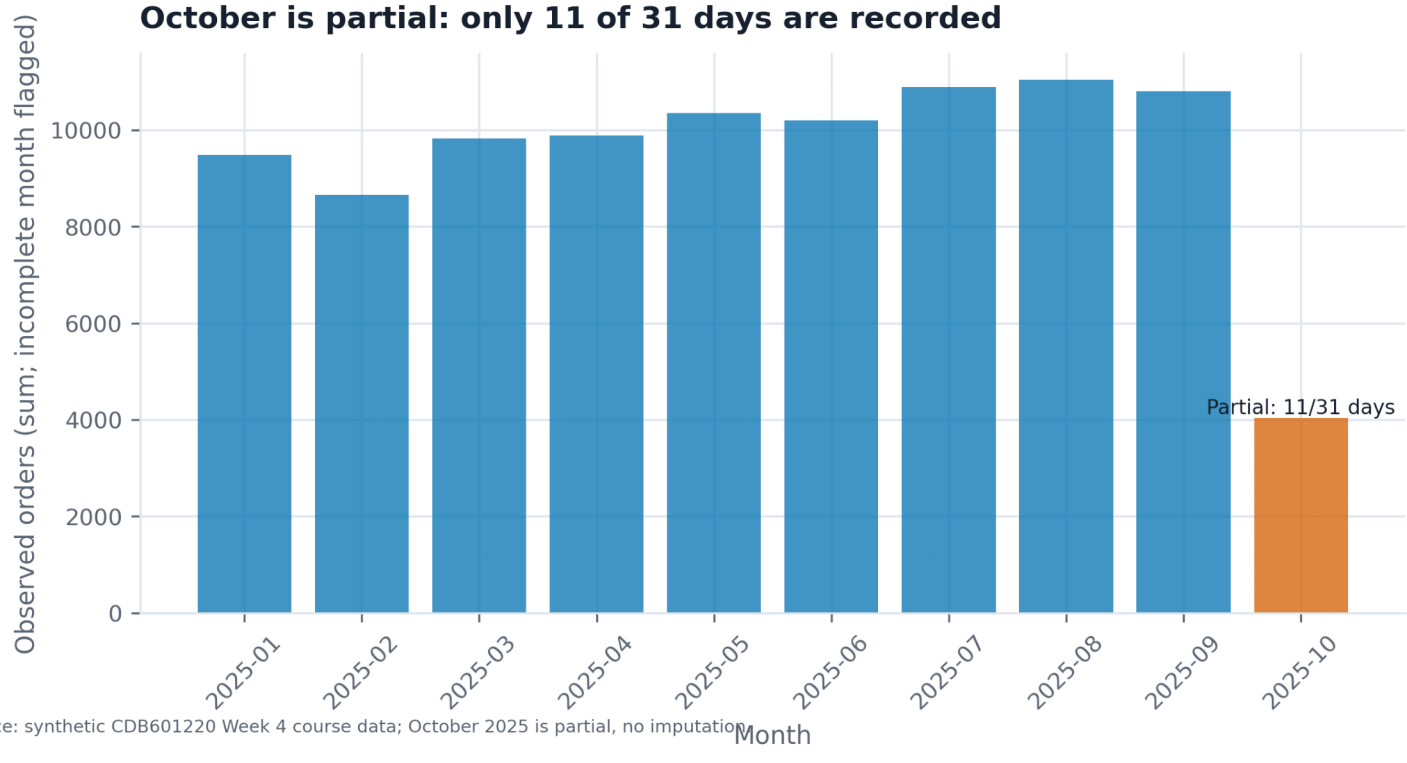
Mechanism: selective time-window framing omits 29 of 36 months. May–November 2025 declines from 158.8 to 92.2 PKR m (-41.9%),
but the full series shows the context and a December recovery to 97.2. The y-axis is already zero-based and n=7 matches the
requested window; neither is the mechanism. The missing context, not axis truncation, makes the claim overbroad.

Case 6 — Published recipe



Source: synthetic CDB601220 Week 4 course data; October contains only observed dates.

Case 6 — Redesign

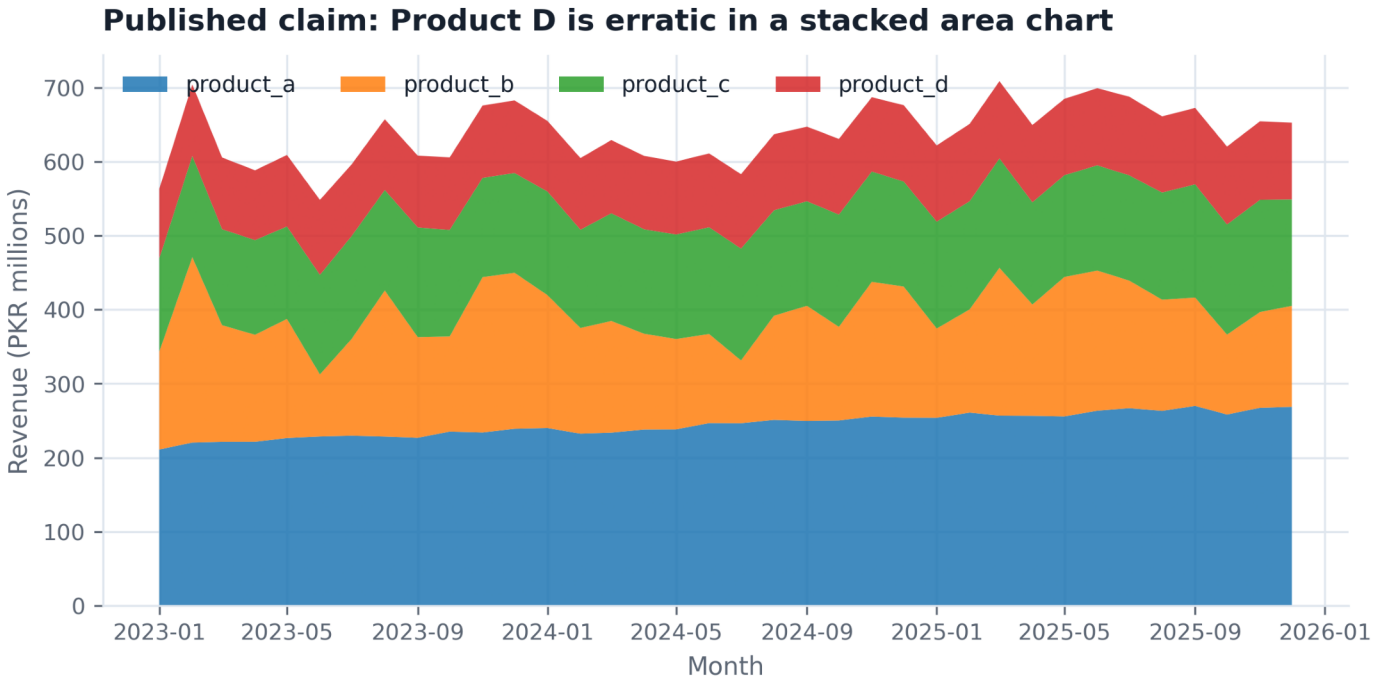


Source: synthetic CDB601220 Week 4 course data; October 2025 is partial, no imputation.

Case 6: diagnosis

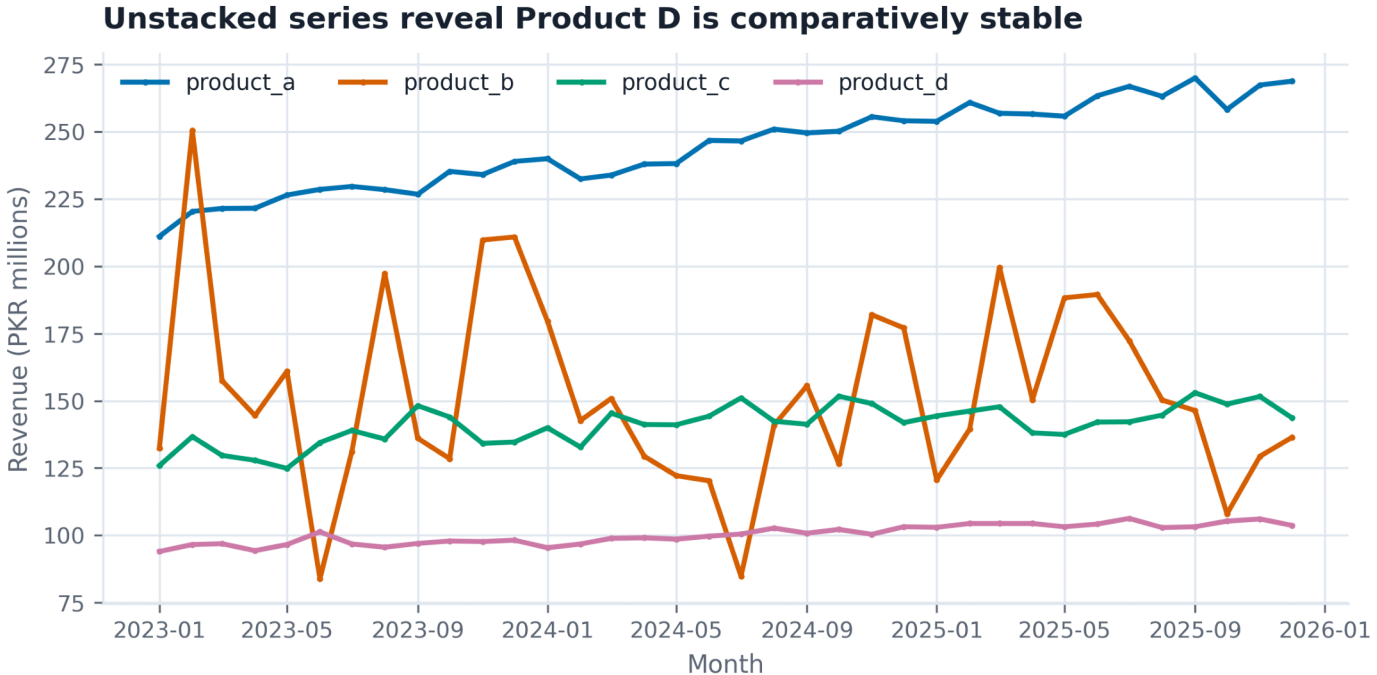
Mechanism: October is a partial month. Only 11 of 31 calendar days are present; the observed sum 4,041 is not comparable with
September's full-month 10,805. A per-day observed rate is 367.4 orders/day in October versus 360.2 in September. Do not project the rate as a full-month total without stating an assumption. The chart is arithmetically correct; its denominator/exposure period differs.

Case 7 — Published recipe



Source: synthetic CDB601220 Week 4 course data; four exclusive product lines.

Case 7 — Redesign

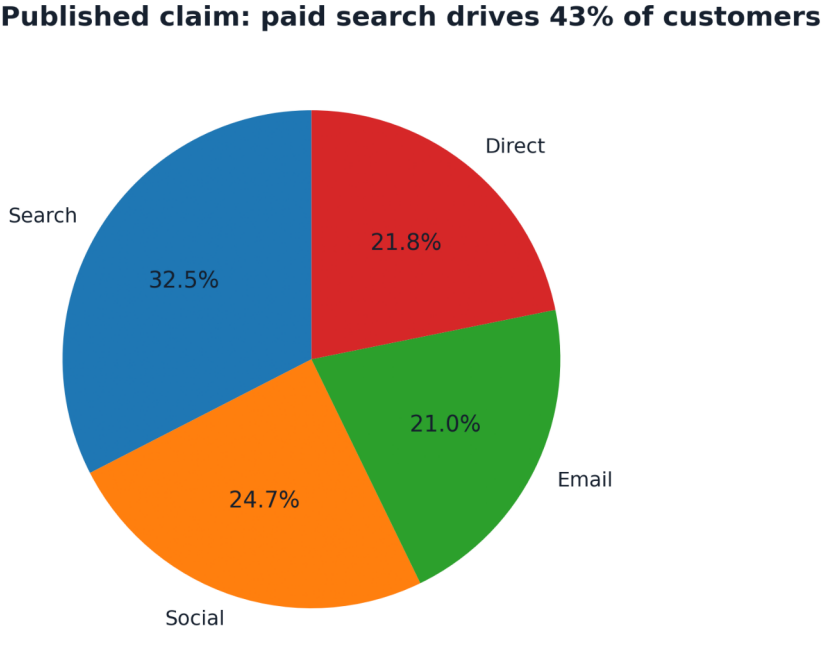


Source: synthetic CDB601220 Week 4 course data; product lines shown separately.

Case 7: diagnosis

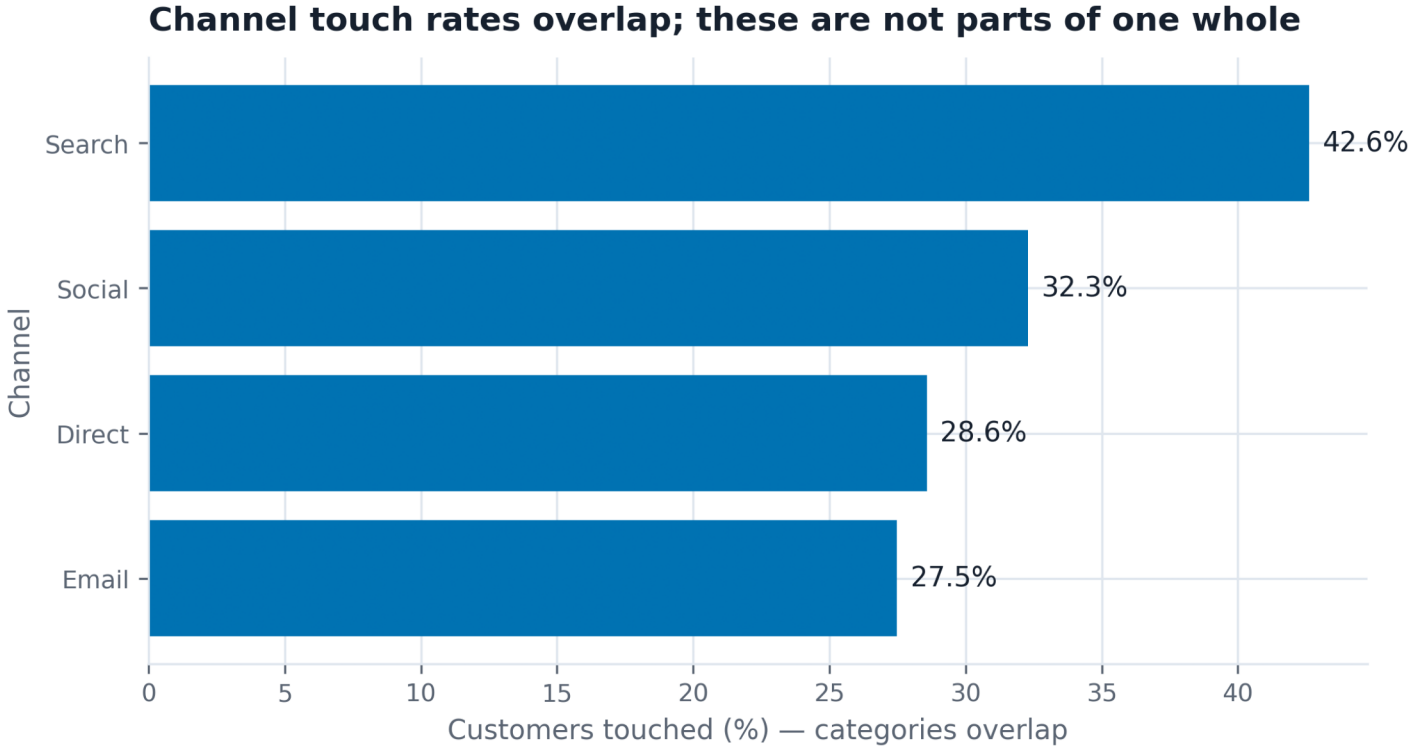
Mechanism: stacking makes the top boundary of Product D equal to the cumulative total A+B+C+D; its visible top edge inherits
movement from all four products. Product D itself ranges only 94.0–106.3 PKR m (SD 3.55), while the total envelope ranges 548.3–708.7. The decoy ‘stacked area is illegal’ is wrong: the dictionary says product lines are exclusive and exhaustive. The problem is reading the top boundary as D's own series.

Case 8 — Published recipe



Source: synthetic CDB601220 Week 4 course data; n=4,000 customers; touch flags may overlap.

Case 8 — Redesign

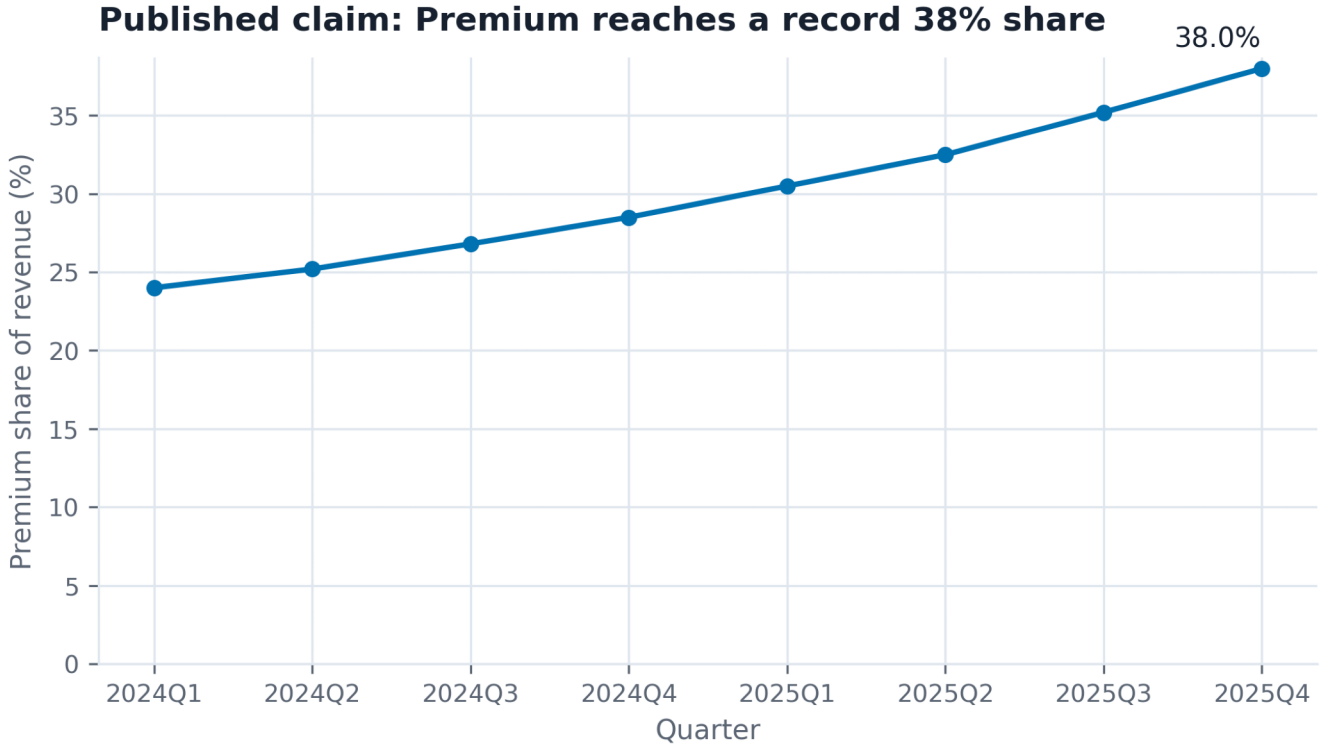


Source: synthetic CDB601220 Week 4 course data; n=4,000 customers; multi-touch customers counted in each applicable channel.

Case 8: diagnosis

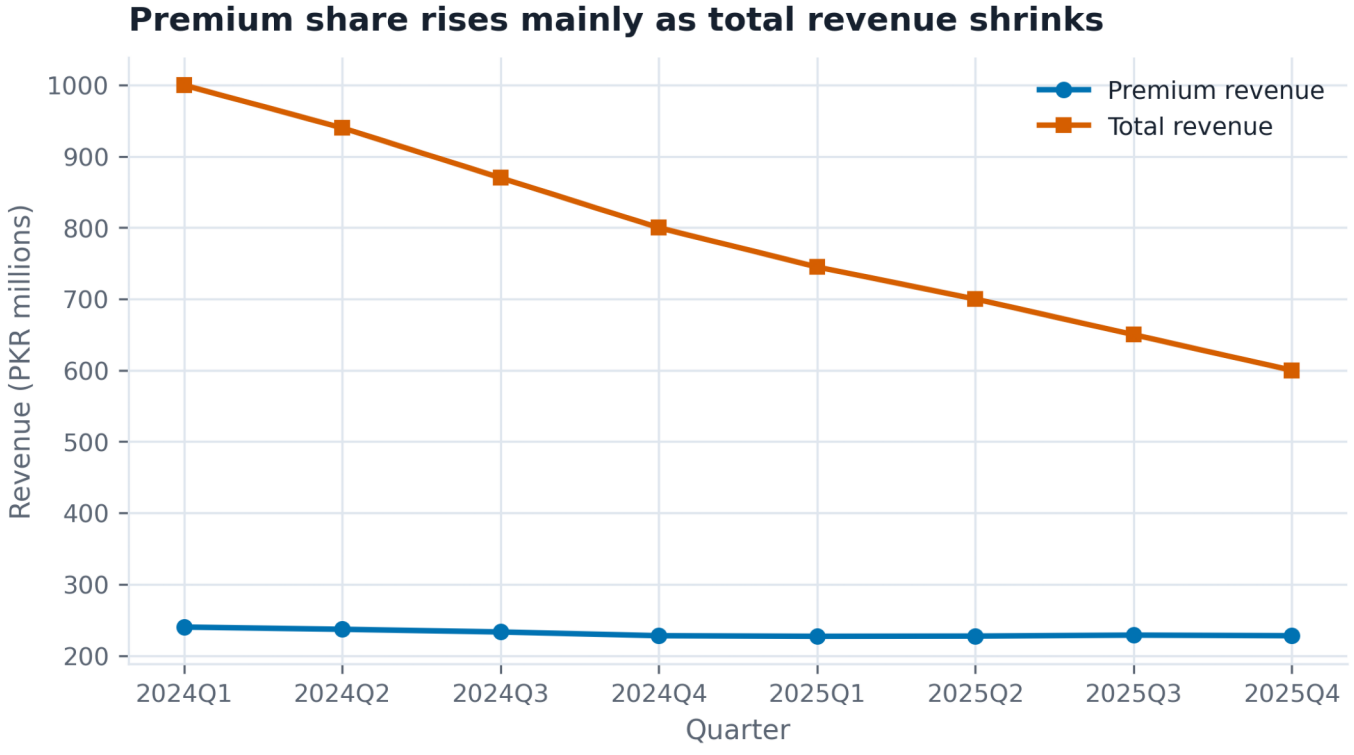
Mechanism: channel flags are non-exclusive; percentages sum to 131.0%, not 100%. 1,105 customers (27.6%) have multiple channel touches and 0 have none. Search touched 42.6% of customers, but this is reach/association, not exclusive attribution or proof that search drove the purchase. The decoy ‘the slices should add to 100 after rounding’ fails: the excess is about 31.0 percentage points, far beyond rounding.

Case 9 — Published recipe



Source: synthetic CDB601220 Week 4 course data; eight quarterly observations.

Case 9 — Redesign



Source: synthetic CDB601220 Week 4 course data; premium and total revenue plotted on same units.

Case 9: diagnosis

Mechanism: denominator contraction raises a ratio even when its numerator does not grow. Premium revenue changes 240.0→228.0
PKR m (-5.0%), while total revenue changes 1000.0→600.0 (-40.0%). Share rises 24.0%→38.0%, but that is not a record in absolute premium revenue; the redesign plots numerator and denominator in the same units.